

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Box 2 i.e. $^{133}_{55}\text{Cs}$		1		1		
	(b)	(i)		Box 3 i.e. the number of neutrons and protons is unbalanced	1			1		
		(ii)		Box 2 i.e. $^0_{-1}\beta$	1			1		
		(iii)		Box 3 i.e. It is a high energy electron	1			1		
	(c)	(i)		80		1		1	1	
		(ii)		Activity will be a quarter / 40 (1) ecf because 2 half-lives / halves again / it is another 30 years (1) so disagree Conclusion must be present to award 2 marks Award 2 marks for: 160 to 80 to 40 so disagree			2	2		
Question 2 total					3	2	2	7	1	0